

cations. It is estimated that over 1,500 surveillance cameras that were being extensively used for security in the units and establishments of the southern naval command were of Chinese origin. The southern naval command, headquartered in Kochi, is the training command of the Indian navy. It has various naval installations in states like Gujarat, Andhra Pradesh, Goa, Odisha, Maharashtra, Tamil Nadu, Kerala and Lakshadweep.

The navy asked for the replacement or relocation of existing CCTV surveillance systems of Hikvision/ Prama Hikvision—the Chinese global market leader of surveillance devices and its Indian subsidiary—from sensitive locations, to be done in a phased manner. To address the security concerns, all naval establishments were asked not to connect these surveillance cameras and the respective storage devices to the internet. "If we continue to instal these Chinese surveillance cameras, we can only blame ourselves. We need to replace them with safe and secure surveillance systems urgently. Otherwise, we are under its (China's) eyes," says Lt Gen. P.C. Katoch (retd), the former director general of information systems in the Indian army.

The Indian intelligence agencies' biggest fear is that the blanket deployment of Chinese-origin CCTVs can, in a worst-case scenario, lead to uninterrupted data leaks to Chinese intelligence. They also suspect that some Chinese firms (working with Indian partners) are sending back data to Beijing through backdoor access to CCTVs—now installed at almost all Indian smart cities, police headquarters, highways and airports, besides central and state government departments, other installations and military establishments. Indeed, Video Security Surveillance (VSS) or CCTV is more than just a camera device being connected to a telecom network. These systems also contain video network recorders, through which all images, stored for later analysis, are connected to a network that can be linked to remote web servers. And that is where the national security risks crop up.

An Indian intelligence official says that modern surveillance cameras come with capabilities that can be switched on remotely when needed—for example, the ability to pick up sounds or read vehicle number plates. He is referring to the system-on-chip (SoC) of a CCTV camera which, with its processor and memory components, operates under software instruction and has apps with different functionalities such as motion and facial recognition embedded in them. The global market for SoCs had been dominated for years by China's HiSilicon, which also supplies them to Hikvision and Dahua, the two firms that are among the world's leading suppliers of CCTVs. Partly owned by the Chinese government, their products are extensively used by the Indian government and are mentioned in the MoD note as potentially dangerous. Once connected to the internet, enormous quantities of visual data can be transmitted back to China. Such disparate material over a variety of sources can potentially be integrated through data fusion to arrive at higher quality intelligence.

It is well-known that data from such surveillance